Daily Calculus & Graphs Drill #1 — Worked Solutions & Investigations

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Section	Topic	Questions	Target time
A	Rapid Recognition	10	2 min 30 sec
B	Manipulation Drills	10	8 min
C	Substitution & Structure	8	10 min
D	Challenge	5	15 min

New toolkit today: Derivative Notation · The Power Rule · Tangents & Normals · The Second Derivative.

Section A Rapid Recognition

1. Find the gradient of the tangent to $y = x^2 - 3x$ at $x = 2$.

Answer: 1

Method: $y' = 2x - 3$, so at $x = 2$ the gradient is $4 - 3 = 1$.

> **Investigate further:** That tangent is $y = x - 4$. Solve $x^2 - 3x = x - 4$ to find where the tangent meets the curve again. Why is the intersection a double root, and why can a tangent to a quadratic never cross it at two distinct points?

2. $f(x) = (x + 1)(x - 3)$. Find $f'(x)$.

Answer: $2x - 2$

Method: Expand first: $(x + 1)(x - 3) = x^2 - 2x - 3$, so $f'(x) = 2x - 2$.

> **Investigate further:** Expand-first is a habit worth testing. Differentiate $f(x) = (x + 1)(x - 3)$ after expanding, then try $f(x) = (x + 1)(x - 3)(x - 5)$. How many terms does the expanded cubic have, and which term disappears when you differentiate? What does that tell you about the information f' keeps?

3. $f(x) = \sqrt{x}$. Find $f'(4)$.

Answer: $\frac{1}{4}$

Method: Write $f(x) = x^{1/2}$, so $f'(x) = \frac{1}{2}x^{-1/2}$ and $f'(4) = \frac{1}{2 \cdot 2} = \frac{1}{4}$.

> **Investigate further:** Find the secant gradient of $y = \sqrt{x}$ between 0 and h. What happens as $h \to 0^+$, and what does that say about the tangent at the origin? Does the secant gradient approach a finite number?

4. $f(x) = \dfrac{1}{x^2}$. Find $f'(x)$.

Answer: $-\frac{2}{x^3}$

Method: Write $f(x) = x^{-2}$, so $f'(x) = -2x^{-3} = -\frac{2}{x^3}$.

> **Investigate further:** State where $f'(x) = -\frac{2}{x^3}$ is positive and where it is negative. Now compare $f(x) = \frac{1}{x}$: there $f'(x) = -\frac{1}{x^2} < 0$ everywhere, yet $f(-1) < f(1)$. What does "decreasing" mean when the domain is not an interval?

5. $f(x) = x^4 - 2x^2$. Find $f''(1)$.

Answer: 8

Method: $f'(x) = 4x^3 - 4x$, then $f''(x) = 12x^2 - 4$, so $f''(1) = 12 - 4 = 8$.

> **Investigate further:** Here $f''(x) = 12x^2 - 4$, so $f''(0) = -4$ but $f''(2) = 44$. What does the sign of f'' say about the gradient as x increases? Find where $f''(x) = 0$ and describe how the curvature changes either side.

6. $f(x) = x^3 - 3x^2 + 4x$. Find $f''(2)$.

Answer: 6

Method: $f'(x) = 3x^2 - 6x + 4$, so $f''(x) = 6x - 6$ and $f''(2) = 12 - 6 = 6$.

> **Investigate further:** $f''(x) = 6x - 6$ changes sign at $x = 1$, so the gradient stops falling and starts rising there. Does that make $x = 1$ a maximum or minimum of f? Evaluate $f'(1)$ and the sign of f' either side before answering.

7. A particle moves along a line with displacement $s = t^3 - 6t^2 + 9t$ metres at time t seconds. Find its velocity at $t = 2$ and state what the sign tells you.

Answer: -3 m/s; moving in the negative direction

Method: $v = \frac{ds}{dt} = 3t^2 - 12t + 9$, so $v(2) = 12 - 24 + 9 = -3$ m/s. The negative sign means the particle is moving in the negative direction.

> **Investigate further:** Solve $v = 0$: the particle is momentarily at rest at $t = 1$ and $t = 3$. Evaluate s at both times and the sign of v between them. Describe the motion in words: where does it turn around, and in which direction does it travel on each interval?

8. Find the equation of the normal to $y = x^2$ at $x = 1$.

Answer: $y = -\frac{1}{2}x + \frac{3}{2}$

Method: At $x = 1$, $y = 1$ and the tangent gradient is 2; the normal gradient is $-\frac{1}{2}$. So $y - 1 = -\frac{1}{2}(x - 1)$, i.e. $y = -\frac{1}{2}x + \frac{3}{2}$.

> **Investigate further:** The normal at $x = 1$ is $y = -\frac{1}{2}x + \frac{3}{2}$. Where does it meet the parabola again? Now find every a for which the normal to $y = x^2$ at $x = a$ passes through the origin; how many such normals are there?

9. $f(x) = \dfrac{x^2 + 4}{\sqrt{x}}$. Find $f'(4)$.

Answer: $\frac{11}{4}$

Method: Split first: $f(x) = x^{3/2} + 4x^{-1/2}$, so $f'(x) = \frac{3}{2}x^{1/2} - 2x^{-3/2}$. At $x = 4$: $\frac{3}{2} \cdot 2 - \frac{2}{8} = 3 - \frac{1}{4} = \frac{11}{4}$.

> **Investigate further:** For $f(x) = \frac{x^2+k}{\sqrt{x}}$, find the value of k for which $f'(4) = 0$. Then evaluate f' at $x = 3$ and $x = 5$ for that k: what does the sign pattern say about the gradient near $x = 4$?

10. $f(x) = ax^3 + bx$, where $f'(1) = 5$ and $f''(1) = 12$. Find a and b.

Answer: $a = 2,\ b = -1$

Method: $f'(x) = 3ax^2 + b$, so $3a + b = 5$. Also $f''(x) = 6ax$, so $6a = 12$, giving $a = 2$ and then $b = 5 - 6 = -1$.

> **Investigate further:** For $f(x) = ax^3 + bx$, show that $f'(-1) = f'(1)$ and $f''(-1) = -f''(1)$ for every a, b. So $f'(1)$ and $f''(1)$ can be chosen freely, but $f'(1)$ and $f'(-1)$ cannot. Which pairs of conditions at $x = 1$ and $x = -1$ determine a and b uniquely?

Section B Manipulation Drills

1. Find the equation of the tangent to $y = x^2 - 4x + 5$ at $x = 3$.

A) $y = 2x + 8$

B) $y = -\frac{1}{2}x + \frac{7}{2}$

C) $y = 2x - 4$

D) $y = 6x - 16$

E) $y = 2x + 5$

Answer: C) $y = 2x - 4$

Method: At $x = 3$, $y = 9 - 12 + 5 = 2$ and $y' = 2x - 4$, so the gradient is 2. Then $y - 2 = 2(x - 3)$, i.e. $y = 2x - 4$. Decoys: A comes from the bracket slip $y - 2 = 2(x + 3)$; B is the normal; D evaluates y' as $2x$ and forgets the -4; E reads the constant term 5 as the y-intercept.

> **Investigate further:** Find the point where the tangent to $y = x^2 - 4x + 5$ is perpendicular to $y = 2x - 4$ ($x = \frac{7}{4}$). Show that the two tangents meet at a point with $y = \frac{3}{4}$. Try another pair of perpendicular tangents to this parabola: what do you notice?

2. Find the equation of the normal to $y = x^2 - 2x$ at $x = 3$.

A) $y = 4x - 9$

B) $y = \frac{1}{4}x + \frac{9}{4}$

C) $y = -4x + 15$

D) $y = -\frac{1}{4}x - \frac{9}{4}$

E) $y = -\frac{1}{4}x + \frac{15}{4}$

Answer: E) $y = -\frac{1}{4}x + \frac{15}{4}$

Method: At $x = 3$, $y = 9 - 6 = 3$ and the tangent gradient is $2x - 2 = 4$, so the normal gradient is $-\frac{1}{4}$. Then $y - 3 = -\frac{1}{4}(x - 3)$, i.e. $y = -\frac{1}{4}x + \frac{15}{4}$. Decoys: A is the tangent; B drops the minus; C negates the gradient without taking the reciprocal; D uses $+3$ instead of -3 for the point's y-coordinate.

> **Investigate further:** Show that the normal meets the parabola again at $x = -\frac{5}{4}$. For a general a, show that the normal at $x = a$ meets $y = x^2 - 2x$ again at $x = 2 - a - \frac{1}{2a-2}$. Why must $a = 1$ be excluded?

3. For $y = x^2 + 2x$, what is the average rate of change between $x = 1$ and $x = 4$?

A) 21

B) 7

C) 4

D) 10

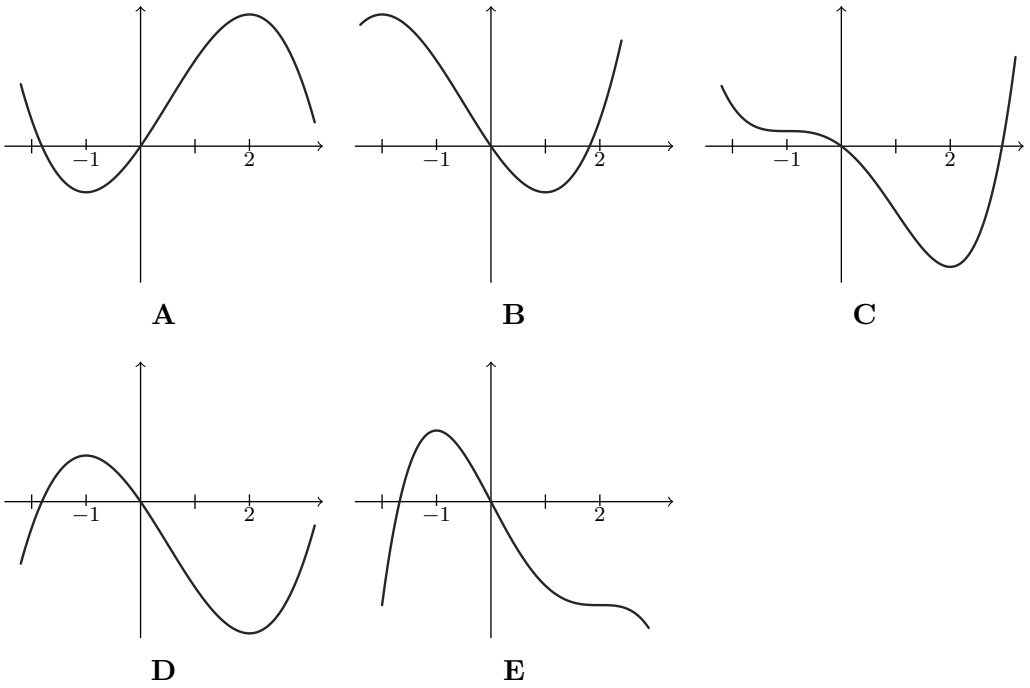
E) $\dfrac{21}{4}$

Answer: B) 7

Method: $f(1) = 3$ and $f(4) = 24$, so the average rate is $\frac{24-3}{4-1} = \frac{21}{3} = 7$. Decoys: A stops at the total change 21; C uses the instantaneous rate $f'(1) = 4$; D uses $f'(4) = 10$; E divides by the upper limit 4 instead of the interval width 3.

> **Investigate further:** Show that the average rate of change of $y = x^2 + 2x$ between $x = p$ and $x = q$ is $p + q + 2$. Which different intervals give the same average rate?

4. The derivative of f satisfies $f'(x) > 0$ for $x < -1$, $f'(-1) = 0$, $f'(x) < 0$ for $-1 < x < 2$, $f'(2) = 0$, and $f'(x) > 0$ for $x > 2$. Which sketch could be $y = f(x)$?



Answer: D

Method: The sign pattern is +, 0, −, 0, +: f rises to a high at $x = -1$, falls to a low at $x = 2$, then rises. Only panel D does this. Decoys: A swaps the high and the low; B puts the flat points at -2 and 1; C has a flat point at -1 where the curve does not turn and only a minimum at 2; E has a maximum at -1, then a flat point at 2 where the curve does not turn and keeps falling.

> **Investigate further:** The correct panel has $f'(x) = (x+1)(x-2)$. Confirm the sign pattern, then explain why a maximum sits where f' changes from $+$ to $-$ and a minimum where it changes from $-$ to $+$. (Day 2 preview: the formal sign chart.)

5. The tangent to $y = x^2$ at $x = a$ passes through $(3, 0)$. Find every possible value of a.

A) $a = 0$ or $a = 6$

B) $a = 6$ only

C) $a = 0$ only

D) $a = 3$

E) $a = -6$ or $a = 0$

Answer: A) $a = 0$ or $a = 6$

Method: The tangent at $x = a$ is $y = 2ax - a^2$. Put $(3, 0)$ on it: $0 = 6a - a^2 = a(6 - a)$, so $a = 0$ or $a = 6$. Decoys: B discards $a = 0$; C discards $a = 6$; D takes the external point's x-coordinate as a; E comes from the sign slip $0 = -6a - a^2$.

> **Investigate further:** For a general point $(p, 0)$, solve $0 = 2ap - a^2$. How many tangents to $y = x^2$ pass through $(p, 0)$ when $p \neq 0$, and what happens when $p = 0$?

6. The normal to $y = x^2$ at $x = a$ is parallel to $y = -\dfrac{x}{4} + 1$. Find a.

A) $a = -2$

B) $a = 2$

C) $a = 4$

D) $a = \dfrac{1}{2}$

E) $a = \dfrac{1}{8}$

Answer: B) $a = 2$

Method: The tangent gradient at $x = a$ is $2a$, so the normal gradient is $-\frac{1}{2a}$. Parallel to $y = -\frac{x}{4} + 1$ means $-\frac{1}{2a} = -\frac{1}{4}$, so $2a = 4$ and $a = 2$. Decoys: A drops the minus; C forgets the factor 2; D reciprocates the answer; E multiplies by 2 instead of dividing.

> **Investigate further:** Show that the normal to $y = x^2$ at $x = a$ is parallel to the tangent at $x = b$ exactly when $ab = -\frac{1}{4}$. What does that say about the tangents at a and b?

7. The tangent to $y = \sqrt{x}$ at $x = 4$ meets the x-axis at

A) $x = 4$

B) $x = 1$

C) $x = -12$

D) $x = -4$

E) $x = \dfrac{9}{2}$

Answer: D) $x = -4$

Method: At $x = 4$, $y = \sqrt{4} = 2$ and $y' = \frac{1}{2\sqrt{x}} = \frac{1}{4}$. The tangent is $y - 2 = \frac{1}{4}(x - 4)$, i.e. $y = \frac{1}{4}x + 1$. Setting $y = 0$ gives $x = -4$. Decoys: A solves $\frac{1}{4}x = 1$ instead of $\frac{1}{4}x = -1$; B reports the y-intercept; C uses $y(4) = 4$ instead of 2; E uses the normal gradient -4 instead of the tangent gradient.

> **Investigate further:** Show that the tangent to $y = \sqrt{x}$ at $x = a$ always meets the x-axis at $x = -a$. Where does it meet the y-axis?

8. For $y = x^3$, find the average rate of change between $x = 1$ and $x = 1 + h$. Simplify your answer, then deduce the instantaneous rate of change at $x = 1$.

Answer: average rate $= 3 + 3h + h^2$; instantaneous rate $= 3$

Method: $f(1) = 1$ and $f(1 + h) = 1 + 3h + 3h^2 + h^3$. The change is $3h + 3h^2 + h^3$, so the average rate is $3 + 3h + h^2$. As $h \to 0^+$, this tends to 3, which is $f'(1) = 3$.

> **Investigate further:** Repeat from $x = a$ to $x = a + h$. Show that the average rate is $3a^2 + 3ah + h^2$, then deduce $f'(a)$.

9. Find the equation of the tangent to $y = x^3 - 3x$ at $x = 2$, and find where that tangent meets the curve again.

Answer: $y = 9x - 16$; again at $(-4, -52)$

Method: At $x = 2$, $y = 8 - 6 = 2$ and $y' = 3x^2 - 3 = 9$. The tangent is $y - 2 = 9(x - 2)$, i.e. $y = 9x - 16$. Solve $x^3 - 3x = 9x - 16$: $x^3 - 12x + 16 = (x - 2)^2(x + 4) = 0$, so the other point is $x = -4$, $y = -52$.

> **Investigate further:** The tangent at $x = 2$ meets the curve again at $x = -4$. Find the tangent at $x = -4$ and check whether it passes through $(2, 2)$. What does that show about tangency as a relation between points?

10. The tangent to $y = x^2 - 2x$ at P is parallel to $y = 2x + 5$. Find the coordinates of P, and the equation of the normal at P.

Answer: $P = (2, 0)$; normal $y = -\frac{1}{2}x + 1$

Method: $y' = 2x - 2$. Parallel to $y = 2x + 5$ means $2x - 2 = 2$, so $x = 2$ and $y = 4 - 4 = 0$. Thus $P = (2, 0)$. The normal gradient is $-\frac{1}{2}$, so $y - 0 = -\frac{1}{2}(x - 2)$, i.e. $y = -\frac{1}{2}x + 1$.

> **Investigate further:** Now find the point on $y = x^2 - 2x$ where the normal is parallel to $y = 2x + 5$ instead. Solve $-\frac{1}{2x-2} = 2$ and interpret the result.

Section C Substitution & Structure

1. $f(x) = \dfrac{x^3 - 8}{x - 2}$ for $x \neq 2$. Find $f'(3)$.

A) 27

B) 8

C) 4

D) 6

E) 19

Answer: B) 8

Method: Factor the difference of cubes: $x^3 - 8 = (x-2)(x^2+2x+4)$, so $f(x) = x^2 + 2x + 4$ for $x \neq 2$. Then $f'(x) = 2x + 2$ and $f'(3) = 8$. Decoys: A differentiates numerator and denominator separately (27/1); C uses the wrong factor $x^2 - 2x + 4$, giving $2x - 2$ and 4; D uses $x^2 + 4$ (the error $a^3 - b^3 = (a-b)(a^2+b^2)$), giving $2x$ and 6; E gives $f(3) = 19$.

> **Investigate further:** $f(x)$ is the chord gradient of $y = x^3$ between 2 and x. Show that $f(x) \to 12$ as $x \to 2$, and explain why that is the derivative of x^3 at 2. Generalise to $\dfrac{x^n - a^n}{x - a}$.

2. The graph of $y = f'(x)$ is a straight line with positive gradient passing through $(1, 0)$. Which could be $f(x)$?

A) $x^2 + 2x - 3$

B) $-x^2 + 2x$

C) $x^3 - 3x$

D) $x^2 - 2x + 5$

E) $2x - 2$

Answer: D) $x^2 - 2x + 5$

Method: f' is a straight line $y = m(x-1)$ with $m > 0$, so f is a quadratic with $f'(1) = 0$ and positive leading coefficient. Only D has $f' = 2x - 2 = 2(x-1)$. Decoys: A has $f(1) = 0$, confusing a root of f with a root of f' (and $f'(1) = 4$); B has $f' = -2(x-1)$, the right line with negative gradient; C has $f'(1) = 0$ but $f' = 3x^2 - 3$ is a parabola, not a line; E is f' itself, not f.

> **Investigate further:** Every $x^2 - 2x + c$ works. Why can't the graph of f' tell you c, and what single extra fact would?

3. A cubic $y = f(x)$ has roots $x = -1$, 2 and 4, and y-intercept -16. Find $f'(0)$.

A) -4

B) 2

C) 4

D) -10

E) -16

Answer: A) -4

Method: Let $f = k(x+1)(x-2)(x-4)$. Then $f(0) = k(1)(-2)(-4) = 8k = -16$, so $k = -2$ and $f = -2x^3 + 10x^2 - 4x - 16$. Hence $f'(x) = -6x^2 + 20x - 4$ and $f'(0) = -4$. Decoys: B assumes $k = 1$, giving $f'(0) = 2$; C uses the factors $(x-1)(x+2)(x+4)$, which force $k = 2$ and $f'(0) = 4$; D takes k times the sum of the roots, $-2(5) = -10$; E gives $f(0) = -16$.

> **Investigate further:** Show that $f'(-1) = k(-1-2)(-1-4) = -30$ without expanding, by taking the chord gradient $\dfrac{f(x) - f(-1)}{x + 1}$ and letting $x \to -1$. Generalise to $f'(p)$ at any root p.

4. For $f(x) = x^3 - 6x^2 + 9x$, for which x are **both** $f'(x) < 0$ and $f''(x) > 0$?

A) $1 < x < 3$

B) $1 < x < 2$

C) $x > 2$

D) $x > 3$

E) $2 < x < 3$

Answer: E) $2 < x < 3$

Method: $f' = 3x^2 - 12x + 9 = 3(x-1)(x-3) < 0$ exactly on $1 < x < 3$, and $f'' = 6x - 12 > 0$ exactly for $x > 2$. Both hold on $2 < x < 3$. Decoys: A ignores f''; B reverses the sign of f'', giving $1 < x < 2$; C ignores f'; D reverses the sign of f', giving $x > 3$.

> **Investigate further:** On $(2, 3)$ the curve is falling but its gradient is increasing. Where is the gradient most negative, and what is that value?

5. The tangents to $y = x^3 + kx^2$ at $x = 1$ and $x = 3$ are parallel. Find k.

A) 6

B) -12

C) -6

D) $-\dfrac{13}{4}$

E) No such k exists

Answer: C) -6

Method: $f' = 3x^2 + 2kx$. Parallel tangents at $x = 1$ and $x = 3$ give $3 + 2k = 27 + 6k$, so $4k = -24$ and $k = -6$. Decoys: A comes from the sign slip $27 - 3 = 6k - 2k$; B differentiates kx^2 as kx, giving $3 + k = 27 + 3k$ and $k = -12$; D equates the y-values, $1 + k = 27 + 9k$, giving $k = -\frac{13}{4}$; E differentiates kx^2 as $2k$, so the k terms cancel and no solution exists.

> **Investigate further:** Prove that the tangents at $x = p$ and $x = q$ ($p \neq q$) are parallel exactly when $p + q = -\dfrac{2k}{3}$. So parallel tangents pair up symmetrically about $x = -\dfrac{k}{3}$.

6. The line $y = 4x + c$ is a tangent to $y = x^3 + x$. Find all possible values of c.

A) $c = \pm 1$

B) $c = \pm 2$

C) $c = -2$ only

D) $c = \pm 3$

E) $c = \pm\dfrac{10\sqrt{3}}{9}$

Answer: B) $c = \pm 2$

Method: $f' = 3x^2 + 1$. A tangent of gradient 4 needs $3x^2 + 1 = 4$, so $x = \pm 1$. The contact points are $(1, 2)$ and $(-1, -2)$. For $y = 4x + c$: $2 = 4 + c$ gives $c = -2$, and $-2 = -4 + c$ gives $c = 2$. Thus $c = \pm 2$.

Decoys: A reports the x-coordinates ± 1; C keeps only $x = 1$; D takes the contact y-values from $y = x^3$ (forgetting the $+x$), giving $\pm 1 - 4(\pm 1) = \mp 3$; E differentiates x as 0, so $3x^2 = 4$, giving $x = \pm\frac{2}{\sqrt{3}}$ and $c = \mp\frac{10\sqrt{3}}{9}$.

> **Investigate further:** For which gradients m does $y = x^3 + x$ have exactly one tangent of gradient m? Exactly two? None?

7. The tangent to $y = \dfrac{4}{x}$ at $x = 3$ meets the axes at P and Q. Find the area of triangle OPQ.

A) 16

B) 0

C) 4

D) 8

E) $\dfrac{25}{2}$

Answer: D) 8

Method: At $x = 3$, $y = \frac{4}{3}$ and $y' = -\frac{4}{x^2} = -\frac{4}{9}$. The tangent is $y - \frac{4}{3} = -\frac{4}{9}(x - 3)$, i.e. $y = -\frac{4}{9}x + \frac{8}{3}$. It meets the axes at $(6, 0)$ and $(0, \frac{8}{3})$, so the area is $\frac{1}{2} \cdot 6 \cdot \frac{8}{3} = 8$. Decoys: A forgets the $\frac{1}{2}$; B uses $+\frac{4}{9}$, which sends the tangent through O and gives area 0; C gives $xy = 3 \cdot \frac{4}{3} = 4$ at the contact point; E differentiates $\frac{4}{x}$ as $-\frac{1}{x^2}$, giving intercepts 15 and $\frac{5}{3}$ and area $\frac{25}{2}$.

> **Investigate further:** Show that the area is 8 wherever the point of contact is, and that the contact point is always the midpoint of PQ. What is the area for $y = \dfrac{k}{x}$?

8. A cubic $f(x) = ax^3 + bx^2 + cx + d$ has tangent $y = 2x + 1$ at $x = 1$, with $f''(1) = 4$ and $f''(0) = -2$. Find $f(2)$.

A) 8

B) 5

C) 6

D) 7

E) Cannot be determined

Answer: A) 8

Method: $f'' = 6ax + 2b$. From $f''(0) = -2$, $b = -1$. From $f''(1) = 4$, $6a - 2 = 4$, so $a = 1$. The tangent at $x = 1$ gives both $f'(1) = 2$ and $f(1) = 3$. Now $f' = 3x^2 - 2x + c$, so $3 - 2 + c = 2$ and $c = 1$; and $f(1) = 1 - 1 + 1 + d = 3$, so $d = 2$. Thus $f = x^3 - x^2 + x + 2$ and $f(2) = 8 - 4 + 2 + 2 = 8$. Decoys: B reads $f(2)$ off the tangent (5); C ignores $f(1) = 3$ and takes $d = 0$ (6); D writes $f'' = 6ax + b$, giving $b = -2$, $a = 1$, $c = 3$, $d = 1$ and $f(2) = 7$; E treats the tangent as one condition instead of two, leaving d undetermined.

> **Investigate further:** The tangent predicts 5 but the true value is 8. Show that $f(1+h) - (3+2h) = h^2(2+h)$, and note that the coefficient of h^2 is $\frac{f''(1)}{2}$. Is that true for every cubic?

Section D Challenge

1. The chord of $y = x^3$ from $x = -1$ to $x = 2$ is parallel to the tangent at $x = c$, where $-1 < c < 2$. Find all such c.

A) $c = \dfrac{1}{2}$

B) $c = -1$ and $c = 1$

C) $c = \sqrt{3}$

D) $c = 1$ only

E) $c = \dfrac{2}{\sqrt{3}}$

Answer: D) $c = 1$ only

Method: Chord gradient: $\dfrac{8-(-1)}{2-(-1)} = \dfrac{9}{3} = 3$. A tangent of gradient 3 needs $3c^2 = 3$, so $c = \pm 1$. The condition $-1 < c < 2$ is strict, so $c = -1$ is excluded and $c = 1$ is the only value. Decoys: A uses the midpoint $\frac{1}{2}$, which is valid only for quadratics; B ignores the strict inequality; C forgets to divide the rise 9 by the run 3, solving $3c^2 = 9$; E uses the chord from $x = 0$, whose gradient is 4, giving $c = \frac{2}{\sqrt{3}}$.

> **Investigate further:** The value $c = -1$ also gives gradient 3. Show the chord's line $y = 3x + 2$ is actually the tangent at $x = -1$, since $x^3 - 3x - 2 = (x+1)^2(x-2)$. Prove that for any quadratic the parallel tangent is always at the midpoint, then find where it is for $y = x^3$ on $[0, h]$.

2. The curves $y = x^2$ and $y = -(x-2)^2$ have $y = 0$ as a common tangent. Find all their other common tangents.

A) $y = 4x - 4$ only

B) $y = -4x - 4$ only

C) $y = 4x - 4$ and $y = -4x - 4$

D) $y = 2x - 1$

E) There are no others

Answer: A) $y = 4x - 4$ only

Method: The tangent to $y = x^2$ at $x = a$ is $y = 2ax - a^2$. The tangent to $y = -(x-2)^2$ at $x = b$ is $y = -2(b-2)x + b^2 - 4$. Matching gradients and intercepts gives $a = 2 - b$ and $-(2-b)^2 = b^2 - 4$, so $2b^2 - 4b = 0$ and $b = 0$ or 2. The root $b = 2$ gives $a = 0$ and the common tangent $y = 0$; the other root $b = 0$ gives $a = 2$ and $y = 4x - 4$. Decoys: B takes the second gradient as $+2(b-2)$, giving $y = -4x - 4$; C assumes mirror symmetry and keeps both; D forces both points of contact to have the same x-coordinate, $2a = -2(a-2)$, giving $a = 1$ and $y = 2x - 1$; E writes the first tangent as $y = 2ax + a^2$, which leaves only $y = 0$.

> **Investigate further:** For $y = x^2$ and $y = -(x-k)^2$, show the common tangents are $y = 0$ and $y = 2kx - k^2$, touching at $x = k$ and $x = 0$ respectively. What happens as $k \to 0$? What if the second curve is shifted to $y = -(x-k)^2 + m$?

3. The curve $y = f(x) = x^3 + bx^2 + cx + d$ has horizontal tangents at $x = -1$ and $x = 3$. Find $f(-1) - f(3)$.

A) -8

B) 64

C) -32

D) Cannot be determined without d

E) 32

Answer: E) 32

Method: Write $f' = 3x^2 + 2bx + c$. Horizontal tangents at $x = -1$ and $x = 3$ give $f' = 3(x+1)(x-3) = 3x^2 - 6x - 9$, so $b = -3$, $c = -9$. Then $f(-1) = 5 + d$ and $f(3) = -27 + d$, so $f(-1) - f(3) = 32$ for every d. Decoys: A forgets the factor 3, using $f' = (x+1)(x-3)$, so $b = -1$, $c = -3$; with the cubic still monic this gives $f(-1) - f(3) = -8$; B assumes the gap is $(3-(-1))^3 = 64$; C computes $f(3) - f(-1) = -32$; D does not notice that d cancels.

> **Investigate further:** Show that for $y = x^3 + bx^2 + cx + d$ with horizontal tangents a distance D apart, the vertical gap is always $\dfrac{D^3}{2}$. Do NOT use integration (that is Day 4): translate so the horizontal tangents are at $x = \pm h$, giving $y = x^3 - 3h^2x + \text{const}$. What changes if the leading coefficient is k?

4. Several tangents to $y = x^3 - 3x$ pass through the point $(2, -2)$. Find the sum of the x-coordinates of all their points of contact.

A) 1

B) 3

C) 2

D) $2 + \sqrt{3}$

E) -3

Answer: B) 3

Method: The tangent at $x = a$ is $y = (3a^2 - 3)x - 2a^3$. Passing through $(2, -2)$ gives $-2 = 2(3a^2 - 3) - 2a^3$, so $a^3 - 3a^2 + 2 = 0$. Since $a = 1$ is a root, $(a-1)(a^2 - 2a - 2) = 0$, giving $a = 1$ and $a = 1 \pm \sqrt{3}$. The sum of the three contact x-coordinates is 3. Decoys: A finds only the horizontal tangent $a = 1$; C keeps only the two roots of the quadratic factor, whose sum is 2; D discards the negative root $1 - \sqrt{3}$ but keeps $a = 1$ and $1 + \sqrt{3}$, giving $2 + \sqrt{3}$; E makes a sign error in the sum of the roots, giving -3.

> **Investigate further:** $(2, 2)$ lies on the curve. Show the equation for a becomes $(a+1)(a-2)^2 = 0$. How many distinct tangents pass through $(2, 2)$, and why is $a = 2$ a double root? (Day 2 preview.) For which points are there exactly three tangents?

5. Let $a \neq 0$.

(a) Show that the tangent to $y = x^3$ at $x = a$ meets the curve again at $x = -2a$.

(b) Show that the gradient of the curve at $x = -2a$ is four times the gradient at $x = a$.

(c) The tangent at $x = -2a$ meets the curve again at $x = b$. Find b, and the gradient there in terms of a.

(d) Explain why the condition $a \neq 0$ is needed.

Answer: $b = 4a$; gradient $48a^2$

Method: (a) The tangent at $x = a$ is $y = 3a^2x - 2a^3$. Subtracting from $y = x^3$ gives $x^3 - 3a^2x + 2a^3$. Factor and expand to verify: $(x-a)^2(x+2a) = (x^2 - 2ax + a^2)(x+2a) = x^3 - 3a^2x + 2a^3$. Since $a \neq 0$, the factor $x + 2a$ gives a second intersection at $x = -2a$. (b) The gradient at $x = -2a$ is $3(-2a)^2 = 12a^2 = 4 \cdot 3a^2$, four times the gradient at $x = a$. (c) Replacing a by $-2a$ in part (a), the tangent at $x = -2a$ meets the curve again at $x = -2(-2a) = 4a$, so $b = 4a$; the gradient there is $3(4a)^2 = 48a^2$. (d) If $a = 0$, the tangent is $y = 0$, and $x^3 = 0$ has only the triple root $x = 0$; there is no second point.

> **Investigate further:** For $y = x^3 + px^2 + qx + r$, show the tangent at $x = a$ meets the curve again at $x = -2a - p$. For which a does it not meet the curve again, and what is special about that point?

Top 5 Patterns — Worksheet #1

1. **Rewrite as powers before differentiating:** $\sqrt{x} = x^{1/2}$, $1/x^2 = x^{-2}$; split or factorise fractions first. *(see A3, A4, A9, B7, C1, C7)*
2. **Tangent at $x = a$:** $y = f'(a)(x-a) + f(a)$. Normal gradient: $-1/f'(a)$. *(see A1, A8, B1, B2, B6, B10)*
3. **Tangency means a repeated root:** substitute the line into the curve, factor out $(x-a)^2$, and the remaining factor gives where the tangent meets the curve again. *(see B9, D1, D5)*
4. **Unknown point of contact:** write the tangent at a general point $x = a$, impose the condition (passes through a point, has a given gradient, is shared by two curves), then solve for a, keeping every root. *(see B5, C6, D2, D4)*
5. **Each condition is one equation:** a value of f, f' or f'' gives one equation, and a tangent line gives two (value and gradient). Count conditions against unknowns. *(see A10, C3, C8, D3)*

Common Traps — Worksheet #1

1. **Differentiating a quotient top and bottom separately:** simplify first. *(see A9, C1)*
2. **Normal gradient:** you must negate AND take the reciprocal; doing only one gives the wrong line. *(see B2, B6)*
3. **Losing roots:** keep obvious roots ($a = 0$ in B5, $a = 1$ in D4) and respect strict inequalities ($c = -1$ is excluded in D1). *(see B5, D1, D4)*
4. **Building f' from turning points:** $f'(x) = 3(x-p)(x-q)$ for a cubic with leading coefficient 1; forgetting the 3 changes every coefficient. *(see C5, D3)*
5. **A tangent line is two conditions, not one:** it fixes both $f(a)$ and $f'(a)$. *(see C8)*

"Simplify first, then differentiate — never the other way round."

Found a mistake or want to contribute a sheet?
Visit the open-source project at github.com/The-CerealDev/speed-maths